THE LONDON SCHOOL OF ECONOMICS AND POLITICAL SCIENCE

Andrews, Kristin, Birch, Jonathan & Sebo, Jeff (2025) Evaluating animal consciousness. *Science, 387*(6736), 822 - 824.
https://doi.org/10.1126/science.adp4990

https://researchonline.lse.ac.uk/id/eprint/127514/

Version: Accepted Version

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Revised draft of *Science* perspective, 12 July 2024

The emerging science of animal consciousness

Kristin Andrews, York University. andrewsk@yorku.ca
Jonathan Birch, London School of Economics and Political Science. j.birch2@lse.ac.uk
Jeff Sebo, New York University. jeffsebo@nyu.edu

Summary sentence: In the absence of a secure theory, the science of consciousness is increasing understanding of animal consciousness through investigating behavioral and neurobiological "markers" of subjective experiences in different species.

Honeybees becoming "pessimistic" after stressful experiences (Bateson et al. 2011), cuttlefish remembering the past and planning for the future (Schnell et al., 2021), cleaner wrasse fish seemingly recognizing themselves in a mirror (Kohda et al., 2023). If scientific reports like these were accepted as evidence of consciousness, significant impact on both research and policy should follow. The neuroscience of consciousness would need to expand its focus beyond on humans and other primates, reaching out to a much wider range of model systems. There should also be changes to how these species are treated, including animal welfare policies that cover all the model species. But note the crucial "if". *Are* these behaviors evidence of consciousness? If so, what conclusions can be drawn when animals perform some behaviors associated with consciousness but not others?

Answering questions about animal consciousness would be easier if we had a secure theory of consciousness, along with a precise definition of 'consciousness' tied to that theory. Unfortunately, theories of consciousness abound, with over 200 theories identified in a recent paper (Kuhn, 2024). While some theories enjoy more empirical support than others, most researchers agree that *all* theories are currently insecure.

Fortunately, we do have many familiar examples of subjectively experienced feelings. These include sensory experiences such as color perception, affective experiences such as bodily pleasure, and vivid imagery during dreams, memories, imagination, or planning. Such examples provide an *initial characterization* of consciousness that allows us to investigate dimensions of conscious experience and to compare these dimensions across species.

However, a key challenge involved with studying animal consciousness without a secure theory is the problem of other minds, widely recognized across world philosophies. John Stuart Mill put the idea simply when he asked, "By what evidence do I know… that the walking and speaking figures which I see and hear, have sensations and thoughts, or in other words, possess Minds?"

↳ we "know" we have minds we just question if others do

Mill answered this question with a kind of analogical argument, identifying similarities between himself and other humans and then inferring that conscious experience is a further similarity: "First, they have bodies like me, which I know in my own case, to be the antecedent condition of feelings; and because, secondly, they exhibit the acts, and outward signs, which in my own case I know by experience to be caused by feelings" (Mill, 1889).

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While this kind of analogical reasoning never delivers certainty, especially when it is applied across species to the question of animal minds, it can still be a valid method for studying animal consciousness when viewed as a kind of *inference to the best explanation*. When humans and other animals perform similar behaviors, and when the best explanation for these behaviors in humans involves conscious experience, then that constitutes evidence—albeit tentative, inconclusive evidence—of conscious experience in other animals, too.

the assumption that we ourselves are conscious gives way to this

Philosophers of science call this kind of inference an "abductive" inference, and it is at the core of the scientific method. Of course, this method can deliver stronger evidence for some "other minds" than for others. When the question is whether other (awake, adult) humans are conscious, this method delivers virtual certainty. However, when the question is whether nonhuman animals are conscious, it delivers less than that. For animals who are highly continuous with us, this method can establish a high degree of confidence that consciousness is present. For animals who are less continuous with us, this method might be able to establish only a "realistic possibility" that consciousness is present – yet this finding can still be significant.

How does the science of animal consciousness determine which continuities are relevant without a secure, agreed upon theory of consciousness? Over the past decade, experts have developed frameworks for conceptualizing the many and varied forms that subjective experiences might take, and have used these frameworks to investigate specific *dimensions* of consciousness (Birch et al. 2020). The first step is to identify a particular dimension of consciousness, such as pain experience, visual experience, or a sense of self. The second step is to identify observable markers of that dimension, and seek empirical evidence that such markers are present (or absent) in the target species. These markers can provide evidence for or against consciousness in that species, either increasing or decreasing our level of confidence as a result. [insert fig?]

↳ these steps directly translate to how we should approach AI consciousness → but there are some features of animals that we may have to verify in AI (animals are sentient, biological/alive, etc)

The degree to which a particular observable marker can increase or decrease our confidence in animal consciousness depends on the context. For instance, linguistic behavior is a marker of particular kinds of conscious thought and emotion in humans, but as chatbots demonstrate, linguistic thought is not, *by itself*, strong evidence of consciousness in nonhuman systems. Awake, adult humans display many interconnected behavioral and neurobiological markers of conscious thought and emotion. When only one of these markers is present, it provides at best very weak evidence of consciousness. But when many of these markers are present together, they jointly provide strong evidence (Bayne et al. 2024).

Careful interpretation is needed when some markers are present and others are absent. On the one hand, when human behavioral markers of consciousness are absent—as with fetuses, infants, and humans who are unresponsive after brain injuries—neurophysiological indicators become more important. On the other hand, when human neurophysiological markers are absent—as with animals with very different brain structures, like octopuses or bees—behavioral markers become more important.

→ in AI as it exists right now, we have the "neuro" by way of interp, but not the "physical"

In these latter cases, a key challenge is to differentiate *conscious* and *unconscious* versions of the relevant behaviors. In the important case of pain, for example, the key is to disentangle experienced pain from nociception, the detection of a noxious stimulus. Nociception is not, itself, a felt experience and it may occur unconsciously. To serve as evidence for pain experience, then, behavior must go beyond mere sensitivity, for example by implicating mechanisms of learning and memory to enable flexible, adaptive, and lasting behavioral change.

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To make progress on this kind of issue, we can begin by distinguishing conscious and unconscious processing in relation to a specific situation (such as encountering tissue damage). We can then identify markers of the conscious version of the behavior in humans (such as behaviors that we perform when, and only when, tissue damage *feels bad*), and we search for species-appropriate versions of these markers in other animals. Clearly, this process leaves room for doubt—and room for improvement. However, these kinds of markers have *face validity*, and they call for further investigation.

Consider markers of pain that researchers have found in different taxa. In a conditioned place preference test conducted on rodents, the individual chooses between two chambers. The initially favored chamber is then paired with a noxious stimulus, while the initially disfavored chamber is paired with an analgesic or local anesthetic. If a human were to switch their preferred chamber in this context, it would be a sign that the noxious stimulus hurt and that the pain relief helped. A similar reversal supplies evidence that the same is true for rodents.

Researchers have found markers of pain in birds as well. Lame broiler chickens demonstrate a greater ability to walk when administered carprofen, an anti-inflammatory drug with analgesic properties. When offered a choice between a normal feed and a feed laced with carprofen, broiler chickens learn to selectively self-administer the analgesic when experiencing leg problems (Danbury et al. 2000). Self-administration of this kind of analgesic supplies evidence that the animal values the kind of pain relief that the analgesic provides.

it seems like this is meant to prove that animals feel pain

What about other vertebrates—reptiles, amphibians, and fishes? Behavioral and neurophysiological markers of experiencing pain have been found in all three. For instance, reptiles show behavioral indicators of pain such as hunched postures and decreased activity (Perry & Nevarez 2018). Amphibians share anatomical and physiological features associated with pain in mammals (Guénette et al. 2013). And fishes not only sense harmful stimuli but also learn to avoid them in the future (Brown & Dory 2019).

Although the search for pain markers in invertebrates is young, when they have been investigated they have often been found. This is particularly true for cephalopod mollusks like octopuses. For instance, octopuses, like rodents, demonstrate conditioned place avoidance. A single trial was enough to produce a shift in preferences, suggesting that acetic acid causes a strongly aversive experience that the lidocaine relieves (Crook 2021).

Pain markers have been found in decapod crustaceans as well. Crabs and shrimps have shown wound-tending and rubbing in response to electric shocks. And when hermit crabs receive electric shocks within a shell, they are less likely to abandon a higher-quality shell than a lower-quality shell, suggesting the shock elicits an unpleasant state that can be weighed centrally against other motivations (Appel and Elwood 2009).

Similar markers are have recently been found in some insects, suggesting that there might be conscious experiences in this vast taxon as well. For example, bees can remember the heat experienced on a heat pad and can trade this off, in their future decision-making, with the sweetness of a sugar reward experienced while on the pad—behavior that goes well beyond nociceptive reflexes (Gibbons et al. 2022).

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Some might object: All of these pain markers pertain to perception, learning, memory, and other such forms of cognition. Is it not possible for an animal to have all of these capacities without it *feeling like anything* from the animal's point of view? The reply to this objection is: Yes, of course this is *possible.* But when we distinguish between conscious and unconscious versions of these functions in humans, identify markers for the conscious versions in humans, and then find these markers in other animals, we can increase our confidence that it feels like something to be these animals, too. → Occam's - why would we assume animals are not conscious if the same markers indicate human consciousness (though feeling pain != conscious)

The idea that consciousness might be widespread across the animal kingdom is increasingly accepted. Many experts now agree on (1) the scientific value of hypothesizing consciousness in other animals, (2) the relevance of pain markers to questions of consciousness, (3) the presence of such markers in all vertebrates and many invertebrates, and, thus, (4) a realistic possibility of consciousness in these animals. The "New York Declaration on Animal Consciousness", launched in April 2024 and signed by 290 experts, captures this changing understanding of consciousness while also communicating ongoing uncertainty (www.nydeclaration.com).

↳ some analog for what framework is necessary for a widely accepted consensus

What does the future hold for this emerging field? As this discussion suggests, the science of animal consciousness is overly reliant on pain markers at present. The top research priority should be to reduce the reliance on this strategy—for both scientific and ethical reasons. There is far more to consciousness than pain. For example, when rats are laughing and bees are playing, is this behavior evidence of a felt emotions of joy? Attempts to develop markers for joy are currently underway (Nelson et al. 2023).

A second priority should be to find new and better methods for disentangling conscious and unconscious processing in animals. In humans and other primates, the study of blindsight is one way to accomplish this goal, but non-invasive techniques (such as backward masking, where a stimulus is followed quickly by a "white noise" stimulus that renders it subliminal) have been influential as well. Future work should investigate how we might be able to translate non-invasive methods to a wider range of animals.

→ we don't share language w/ animals so we must develop these methods - LLMs make this easier

As the science of animal consciousness evolves, the evidence regarding consciousness in different taxa will evolve as well, supporting higher or lower degrees of confidence that it feels like something to be different animals. The idea that there is a "realistic possibility" of consciousness in all vertebrates and many invertebrates may eventually be replaced by more confident language, one way or the other. But for as long as the evidence remains limited and mixed, it is important to keep an open mind and strive to learn more.

The wider lesson is: We cannot simply assume that all forms of conscious experience are uniquely human. Instead, we must take seriously the idea that some dimensions of conscious experience exist in other animals too, and we must hypothesize relevant similarities between our experiences and theirs. Yes, we must be cautious and tentative when making attributions of consciousness across species. But when these working hypotheses lead to fruitful research programs, abductive reasoning supports the conclusion that the similarity is real.

At this early stage in consciousness science, it is important to accept the absence (for now) of a secure theory or a precise definition, along with a lack of proof or certainty. However, while proof or certainty are not achievable goals for this field (for now), they are also not the only legitimate scientific goals: *Evidence* is also a legitimate goal, and an achievable one. Seeking

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evidence of the different dimensions of consciousness across species will help us to build towards better theories of consciousness—and a better understanding of the tree of life.

→ to research/address separately: what role does their consciousness have to play in the ethics of our relationship with/treatment of animals?

Acknowledgements

Thanks to Toni Sims for research support, to Sofia Fogel for project support, and to the initial signatories of the 2024 New York Declaration on Animal Consciousness for useful feedback and discussion on a draft.

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Figure

The marker-based approach to questions of consciousness in other animals

Although no marker conclusively proves the presence of consciousness in nonhuman animals, many markers can substantially raise the probability that consciousness exists.

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